29.1: Plants evolved from green algae

Today, there are more than 325,000 known plant species. Although a few plant species returned to aquatic habitats during their evolution, most present-day plants live on land. In this text, we distinguish plants from algae, which are photosynthetic protists.

Plants enabled other life-forms to survive on land. For example, plants supply oxygen and are a key source of food for terrestrial animals. Also, by their very presence, plants such as the trees of a forest physically create the habitats required by animals and many other organisms.

As you read in Concept 28.4, green algae called charophytes are the closest relatives of plants. We'll begin with a closer look at the evidence for this relationship.

Evidence of Algal Ancestry

Many key traits of plants are found in some algae. For example, plants are multicellular, eukaryotic, photosynthetic autotrophs, as are brown, red, and certain green algae. Plants have cell walls made of cellulose, and so do green algae, dinoflagellates, and brown algae. And chloroplasts with chlorophylls a and b are present in green algae, euglenids, and a few dinoflagellates, as well as in plants.

However, the charophytes are the only present-day algae that share certain distinctive traits with plants, suggesting that they are the closest living relatives of plants. For example, the cells of both plants and charophytes have distinctive circular rings of proteins embedded in the plasma membrane **(Figure 29.2)**; these protein rings synthesize the cellulose found in the cell wall. In contrast, noncharophyte algae have linear sets of proteins that synthesize cellulose. Likewise, in species of plants that have flagellated sperm, the structure of the sperm closely resembles that of charophyte sperm. Analyses of nuclear, chloroplast, and mitochondrial DNA also support the close relationship between plants and charophytes.

Figure 29.2

Rings of cellulose-synthesizing proteins.

Rings of cellulose-synthesizing proteins on a 20 nanometer scale.

(inverted-contrast TEM)

More specifically, a recent analysis of nearly 900 protein-coding nuclear genes found that charophytes in the clade Zygnematophyceae—such as *Zygnema*—are the closest living relatives of plants **(Figure 29.3)**. Although this evidence shows that plants arose from within a group of charophyte algae, it does not mean

that plants are descended from these living algae. Even so, present-day charophytes may tell us something about the algal ancestors of plants.

Figure 29.3

Zygnema, an alga closely related to plants.

Zygnema, an alga in a tube-like structure. Scale = 20 micrometers.

Zygnema is a filamentous pond alga in the Zygnematophyceae, the group of charophytes that is most closely related to plants.

Adaptations Enabling the Move to Land

Many species of charophyte algae inhabit shallow waters around the edges of ponds and lakes, where they are subject to occasional drying. In such environments, natural selection favors individual algae that can survive periods when they are not submerged. In charophytes, a layer of a durable polymer called **sporopollenin** prevents exposed zygotes from drying out. A similar chemical adaptation is found in the tough sporopollenin walls that encase plant spores.

The accumulation of such traits by at least one population of charophyte algae (now extinct) enabled their descendants—the first plants—to live permanently above the waterline. This ability opened a new frontier: a terrestrial habitat that offered enormous benefits. The bright sunlight was unfiltered by water and plankton; the atmosphere offered more plentiful carbon dioxide than did water; and the soil by the water's edge was rich in some mineral nutrients. But these benefits were accompanied by challenges: a relative scarcity of water and a lack of structural support against gravity. (To appreciate why such support is important, picture how the soft body of a jellyfish (jelly) collapses when waves strand it on a beach.) Plants diversified as new adaptations arose that enabled them to thrive despite these challenges.

Today, what adaptations are unique to plants? The answer depends on where you draw the boundary dividing plants from algae **(Figure 29.4)**. Since the placement of this boundary is the subject of ongoing debate, this text uses a traditional definition that equates the kingdom Plantae with embryophytes (plants with embryos). In this context, let's now examine the derived traits that separate plants from their closest algal relatives.

Figure 29.4

Three possible "plant" kingdoms, the clades Viridiplantae, Streptophyta, and Plantae.

Tree diagram for the evolution of algae, chlorophytes, charophytes and embryophytes from ancestral alga.

Tree diagram for the evolution of algae, chlorophytes, charophytes and embryophytes from ancestral alga with limbs represented as a list with the top of the list. Common ancestors are identified at branch points if labeled; if not, branch point is noted.

Branch point 1: Ancestral alga Red algae.

Branch point 2 (Viridiplantae). Chlorophytes.

Branch Point 3 (Streptophyta). Other charophytes.

Branch point 4. Closest charophyte relative. Embryophytes (Plantae).

Derived Traits of Plants

Several adaptations that facilitate survival and reproduction on dry land emerged after plants diverged from their algal relatives. Examples of such traits that are found in plants but not in charophyte algae include the following:

- **Alternation of generations.** This type of life cycle, consisting of multicellular forms that give rise to each other in turn, is described in **Figure 29.5**.

Figure 29.5

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Exploring Alternation of Generations
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Exploring derived traits of land plants – alternation of generations, multicellular dependent embryos, walled spores produced in sporangia, multicellular gametangia, and apical meristems.

Charophyte algae lack the key traits of plants described in this figure: alternation of generations; multicellular, dependent embryos; walled spores produced in sporangia; multicellular gametangia; and apical meristems. This suggests that these traits were absent in the ancestor common to plants and charophytes but instead evolved as derived traits of plants. Not every plant exhibits all of these traits; certain lineages of plants have lost some traits over time. Five steps of alternation of generations from the gametophyte producing haploid gametes (mitosis), zygote formation (fertilization), sporophyte formation (mitosis), haploid spores (meiosis), and gametophytes (mitosis), and the cycle continues. The life cycles of all plants alternate between two generations of distinct multicellular organisms: gametophytes and sporophytes. As shown in the diagram below (using a fern as an example), each generation gives rise to the other, a process that is called alternation of generations. This type of reproductive cycle evolved in various groups of algae but does not occur in the charophytes, the algae most closely related to plants. Take care not

to confuse the alternation of generations in plants with the haploid and diploid stages in the life cycles of other sexually reproducing organisms (see Figure 13.6). Alternation of generations is distinguished by the fact that the life cycle includes both multicellular haploid organisms and multicellular diploid organisms. The multicellular haploid gametophyte (gamete-producing plant) is named for its production by mitosis of haploid gametes—eggs and sperm—that fuse during fertilization, forming diploid zygotes. Mitotic division of the zygote produces a multicellular diploid sporophyte (spore-producing plant). Meiosis in a mature sporophytes produces haploid spores, reproductive cells that can develop into a new haploid organism without fusing with another cell. Mitotic division of the spore cell produces a new multicellular gametophyte, and the cycle begins again. Image: 1. The gametophyte (n) produces haploid gametes by mitosis. Gamete (n) from another plant is introduced and 2. The two gametes unite (fertilization) and form diploid zygote. 3.The zygote (2 n) develops into a multicellular diploid sporophyte by mitosis. 4. The sporophyte produces unicellular haploid spores by meiosis. 5. The spores (n) develop into multicellular haploid gametophytes and the cycle begins again. Key: haploid (n), diploid (2 n).

MAKE CONNECTIONS *Review sexual life cycles in*Figure 13.6*. Identify which type of sexual life cycle has alternation of generations, and summarize how it differs from other life cycles.*

*For suggested answer, see*Appendix A.*

- **Walled spores produced in sporangia.** The sporophyte stage of the plant life cycle has multicellular organs called **sporangia** (singular, *sporangium*) that produce spores **(Figure 29.6)**. The polymer sporopollenin makes the walls of these spores resistant to harsh environments, enabling plant spores to be dispersed through dry air without harm.

Figure 29.6

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Sporophytes and sporangia of a moss in the genus *Mnium*.
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Sporophytes and sporangia of Sphagnum (a moss).

Each of the many spores produced by a sporangium is encased by a durable, sporopollenin-enriched wall.

- **Apical meristems**. Plants also differ from their algal relatives in having ********apical meristems**, localized regions of cell division at the tips of roots and shoots **(Figure 29.7)**. Apical meristem cells can divide throughout the plant's life, enabling its roots and shoots to elongate, thus increasing the plant's exposure to environmental resources.

Figure 29.7

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Apical meristem of a plant shoot.
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Apical meristems of plant roots and shoots. The LMs are longitudinal sections at the tips of a root and shoot.

Cells produced by apical meristems differentiate into the outer epidermis, which protects the body, and various types of internal tissues, such as those in leaves (LM).

Additional derived traits that relate to terrestrial life have evolved in many plant species. For example, the epidermis in many species has a covering, the **cuticle**, which consists of wax and other polymers. Permanently exposed to the air, plants run a far greater risk of desiccation (drying out) than do their algal relatives. The cuticle acts as waterproofing, helping prevent excessive water loss from the aboveground plant organs, while also providing some protection from microbial attack. Most plants also have specialized pores called **stomata** (singular, *stoma*), which support photosynthesis by allowing the exchange of CO2 and O2 between the outside air and the plant (see Figure 10.3). Stomata are also the main avenues by which water evaporates from the plant; in hot, dry conditions, the stomata close, minimizing water loss.

The earliest plants lacked true roots and leaves. Without roots, how did these plants absorb nutrients from the soil? Fossils dating from 420 million years ago reveal an adaptation that may have aided early plants in nutrient uptake: They formed symbiotic associations with fungi. We'll explore these associations, called *mycorrhizae*, and their benefits to both plants and fungi in more detail in Concept 31.1. For now, the main point is that mycorrhizal fungi form extensive networks of filaments through the soil and transfer nutrients to their symbiotic plant partner. This benefit may have helped plants without roots to colonize land.

The Origin and Diversification of Plants

The algae most closely related to plants include many unicellular species and small colonial species. Since it is likely that the first plants were similarly small, the search for the earliest fossils of plants has focused on the microscopic world. As mentioned earlier, microorganisms colonized land as early as 3.2 billion years ago. But the microscopic fossils that document life on land changed dramatically 470 million years ago with the appearance of spores from early plants.

What distinguishes these spores from those of algae or fungi? One clue comes from their chemical composition, which matches the composition of plant spores today but differs from that of the spores of other organisms. In addition, the walls of these ancient spores have structural features that today are found only in the spores of certain plants (liverworts). And in rocks dating to 450 million years ago, researchers have discovered similar spores embedded in plant cuticle material that resembles spore-bearing tissue in living plants **(Figure 29.8)**.

Figure 29.8

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Ancient plant spores and tissue.
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Ancient plant spores and tissue.

(colorized SEMs)

Part a. Fossilized spores. The chemical composition and the physical structure of the walls of these 450-million-year-old spores match those found in certain plants today.

Part b. Fossilized sporophyte tissue. The spores in part a were embedded in tissue that appears to be from plants. Scale = 50 micrometers.

Fossils of larger plant structures, such as the *Cooksonia* sporangium in **Figure 29.9**, date to 425 million years ago, which is 45 million years after the appearance of plant spores in the fossil record. While the precise age (and form) of the first plants has yet to be discovered, those ancestral species gave rise to the vast diversity of living plants. **Table 29.1** provides basic information about the ten extant phyla. (Extant lineages are those that have surviving members.) As you read the rest of this section, look at Table 29.1 together with **Figure 29.10**, which reflects a view of plant phylogeny that is based on plant anatomy, biochemistry, and genetics.

Figure 29.9

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*Cooksonia* sporangium fossil.
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Mushroom-shaped fossil of Cooksonia sporangium; scale = 0.3 mm.

With the exception of the shape of its sporangia, *Cooksonia's* overall form was very similar to the early plant shown in Figures 29.1 and 29.16.

Table 29.1

Ten Phyla of Extant Plants

A table titled Table 29.1 Ten phyla of extant plants illustrates common name and number of known species.

	Common Name	Number of Known Species

Nonvascular Plants (Bryophytes)		
Phylum Hepatophyta	Liverworts	9,000
Phylum Bryophyta	Mosses	13,000
Phylum Anthocerophyta	Hornworts	225
Vascular Plants		
Seedless Vascular Plants		
Phylum Lycophyta	Lycophytes	1,200
Phylum Monilophyta	Monilophytes	12,000
Seed Plants		
Gymnosperms		
Phylum Ginkgophyta	Ginkgo	1
Phylum Cycadophyta	Cycads	350
Phylum Gnetophyta	Gnetophytes	75
Phylum Coniferophyta	Conifers	600
Angiosperms		
Phylum Anthophyta	Flowering plants	290,000

Figure 29.10

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Highlights of plant evolution.
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Tree diagram for the highlights of plant evolution showing land plants (nonvascular plants, seedless vascular plants, and seed plants) from ancestral green alga.

The phylogeny shown here illustrates a leading hypothesis about the relationships between plant groups.

Tree diagram for the highlights of plant evolution of land plants (nonvascular plants, seedless vascular plants, and seed plants) from ancestral green alga with limbs represented as a list.

Branch point 1: Ancestral green alga (origin of land plants (about 470 million years ago (million years ago)). Liverworts, nonvascular plants (bryophytes).

Branch point 2 (about 460 million years ago). Mosses - nonvascular plants (bryophytes).

Branch Point 3 (about 440 million years ago). Hornworts - nonvascular plants (bryophytes)

Branch Point 4, origin of vascular plants (about 425 million years ago). Lycophytes (club mosses, spike mosses, quillworts) - seedless vascular plants.

Branch point 5 (about 390 million years ago). Monilophytes (ferns, horsetails, whisk ferns), seedless vascular plants.

Branch point 6, origin of seed plants (about 360 million years ago). Gymnosperms. Angiosperms.

MAKE CONNECTIONS *The figure identifies which lineages are plants, nonvascular plants, vascular plants, seedless vascular plants, and seed plants. Which of these categories are monophyletic, and which are paraphyletic? Explain. (See**Figure 26.10**to review these terms.)*

*For suggested answer, see*Appendix A.**

One way to distinguish groups of plants is whether or not they have an extensive system of **vascular tissue**, cells joined into tubes that transport water and nutrients throughout the plant body. Most present-day plants have a complex vascular tissue system and are therefore called **vascular plants**. Plants that do not have an extensive transport system—liverworts, mosses, and hornworts—are described as “nonvascular” plants, even though some mosses do have simple vascular tissue. Nonvascular plants are often informally called **bryophytes** (from the Greek *bryon*, moss, and *phyton*, plant). Although the term *bryophyte* is commonly used to refer to all nonvascular plants, molecular studies and morphological analyses of sperm structure have concluded that bryophytes do not form a monophyletic group (a clade).

Vascular plants, which form a clade that comprises about 93% of all extant plant species, can be categorized further into smaller clades. Two of these clades are the **lycophytes** (the club mosses and their relatives) and the **monilophytes** (ferns and their relatives). The plants in each of these clades lack seeds, which is why collectively the two clades are informally called **seedless vascular plants**. However, notice in Figure 29.10 that, like bryophytes, seedless vascular plants do not form a clade.

A group such as the bryophytes or the seedless vascular plants represents a collection of organisms that share key biological features. It can be informative to group organisms according to their features, such as having a vascular system but lacking seeds. But members of such a group, unlike members of a clade, do not necessarily share the same ancestry. For example, even though monilophytes and lycophytes are all seedless vascular plants, monilophytes share a more recent common ancestor with seed plants. As a result, we would

expect monilophytes and seed plants to share key traits not found in lycophytes—and they do, as you'll read in Concept 29.3.

A third clade of vascular plants consists of seed plants, which represent the vast majority of living plant species. A **seed** is an embryo packaged with a supply of nutrients inside a protective coat. Seed plants can be divided into two groups, gymnosperms and angiosperms, based on the absence or presence of enclosed chambers in which seeds mature. **Gymnosperms** (from the Greek *gymnos*, naked, and *sperm*, seed) are known as "naked seed" plants because their seeds are not enclosed in chambers. Living gymnosperm species, the most familiar of which are the conifers, form a clade. **Angiosperms** (from the Greek *angion*, container) are a huge clade consisting of all flowering plants; their seeds develop inside chambers that originate within flowers. Nearly 90% of living plant species are angiosperms.

Note that the phylogeny depicted in Figure 29.10 focuses only on the relationships between extant plant lineages. Paleobotanists have also discovered fossils belonging to extinct plant lineages. As you'll read later in the chapter, these fossils can reveal intermediate steps in the emergence of plant groups found on Earth today.